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AI CAREERS FOR WOMEN

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Hospital Patient Records And Health Metrics Dashboard

Project Domain

Excel / Power BI

A Collaborative Initiative

equipping students with practical skills in AI, Data Analytics, and Business Intelligence

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3. Problem Statement

The healthcare and hospital management domain generates a large volume of patient and operational data every day, including admissions, diagnoses, treatments, laboratory reports, bed allocation, and billing records. In many hospitals, this data is stored across multiple Excel sheets, paper records, or disconnected systems, leading to inconsistencies, duplication, missing values, and difficulty in accessing real-time information. These data-related challenges make it hard to prepare accurate reports and monitor hospital performance effectively.

This problem is significant in real-world scenarios because delayed or inaccurate data can negatively impact patient care, resource planning, and hospital efficiency. Administrators often depend on manual reporting methods, which are time-consuming and error-prone, making it difficult to track key health metrics such as bed occupancy rate, patient flow, departmental workload, and treatment outcomes. The lack of centralized dashboards and visual analytics creates a clear gap in timely decision-making and performance monitoring.

To address this inefficiency, the project leverages Power BI to clean, integrate, and visualize hospital patient records through interactive dashboards and KPIs. This solution provides centralized data management, real-time insights, automated reporting, and easy-to-understand visualizations. Hospital administrators, doctors, and management staff benefit from faster decision-making, improved operational efficiency, better resource allocation, and enhanced patient care quality.

1. BACKGROUND OF THE DOMAIN AND DATA-RELATED CHALLENGES

The healthcare and hospital management domain generates a large amount of data every day, including patient admissions, diagnoses, treatments, laboratory reports, bed allocation, discharge summaries, and billing records. In many hospitals, this information is stored in multiple formats such as Excel sheets, paper records, or different disconnected systems.

Due to this fragmented storage, hospitals often face several data-related challenges such as data duplication, missing values, inconsistencies, and difficulty in accessing real-time information. These issues make it difficult for hospital staff and administrators to maintain accurate records and analyze operational performance effectively.

2. SIGNIFICANCE OF THE PROBLEM IN A REAL-WORLD CONTEXT

This problem is highly significant in real-world healthcare environments because timely and accurate data is critical for patient care and hospital management. When hospital data is scattered or poorly organized, it becomes difficult to track important performance indicators such as bed occupancy rates, patient flow, department workloads, and treatment outcomes.

Many hospitals still rely on manual reporting methods, which are time-consuming and prone to human error. As a result, decision-makers may not have access to up-to-date insights, which can negatively affect resource planning, operational efficiency, and patient care quality.

3. GAP OR INEFFICIENCY THE PROJECT AIMS TO ADDRESS

The key gap identified in this domain is the lack of a centralized system for analyzing and visualizing hospital data. Without proper dashboards or automated reporting tools, hospital administrators struggle to quickly interpret data and make informed decisions.

This project aims to address this inefficiency by using Power BI to clean, integrate, and visualize hospital patient records. The project transforms raw hospital data into interactive dashboards, KPIs, and visual reports, enabling easier monitoring of hospital operations and performance.

4. STAKEHOLDERS AND VALUE DELIVERED

The proposed solution benefits several stakeholders, including hospital administrators, doctors, healthcare staff, and management teams. By providing centralized data management and real-time analytics, the dashboard helps them quickly understand key metrics and trends.

The value delivered by this solution includes:

- Faster and more accurate decision-making
- Improved hospital operational efficiency
- Better resource allocation and planning
- Reduced manual reporting workload
- Enhanced patient care and service quality

4. Objectives

• To collect, organize, and clean hospital patient records from multiple sources to ensure accurate, consistent, and reliable data for analysis.

• To integrate and centralize the data into a unified system using Microsoft Power BI for better accessibility and management.

• To design interactive dashboards and visual reports that display key health metrics such as admissions, discharges, bed occupancy rate, treatment outcomes, and departmental performance.

• To enable real-time monitoring of hospital operations through automated KPIs and dynamic visualizations for quick insights.

• To support hospital administrators and medical staff in making data-driven decisions that improve resource allocation, workflow efficiency, and service quality.

• To reduce manual reporting efforts, minimize errors, and enhance overall patient care and operational effectiveness.

• To implement data filtering and slicer options that allow users to analyze records department-wise, doctor-wise, and date-wise for deeper insights.

• To identify trends and patterns in patient admissions, disease occurrence, and treatment success rates to support strategic hospital planning.

5. Methodology / Implementation Plan

Provide a simple overview of how the project will be carried out. The following five-phase workflow applies to all projects under this fellowship program.

#	Phase	Description
1	**Data Collection & Exploration**	Gather hospital patient records from multiple sources such as Excel sheets, databases, or hospital systems and understand the structure, data types, and key attributes. Perform initial exploration to identify patterns and issues.
2	**Data Cleaning & Preprocessing**	Remove duplicates, handle missing values, correct errors, standardize formats, and transform raw data into a structured and consistent dataset suitable for analysis.

3	**Analysis & AI Integration**	Analyze patient trends, admissions, discharges, and health metrics. Apply basic analytics and AI-based insights such as forecasting, pattern detection, or KPI calculations to support decision-making.
4	**Visualization / Output**	Create interactive dashboards, charts, and KPIs in Microsoft Power BI to present insights clearly through graphs, filters, and slicers for real-time monitoring.
5	**Documentation & Submission**	Prepare project reports, explain methodology, include screenshots of dashboards, summarize findings, and submit all required documents and deliverables..

6. Resources and Timeline

A. Hardware & Software Requirements

Minimum hardware needed to run the system:

- Processor: Intel i3/i5 or higher
- RAM: 4–8 GB minimum
- Storage: 256 GB or above
- System: Desktop/Laptop
- Internet connection for dashboard updates

Software Requirements

Core Software:

- Microsoft Excel – Data cleaning and preprocessing
- Microsoft Power BI – Dashboard creation and visualization

Supporting Software:

- Windows 10/11 Operating System
- Microsoft Office Suite
- Web browser (Chrome/Edge)
- Optional: Database systems for future expansion

B. Dataset

• Dataset generated and downloaded using **Microsoft Copilot** to simulate real-world hospital patient records
• Contains 500+ patient entries with demographic, admission, diagnosis, treatment, billing, and discharge details
• Includes multiple data types such as numerical, categorical, date, and text fields for comprehensive analysis
• Cleaned and preprocessed before importing into **Microsoft Power BI**
• Used to create KPIs, reports, and interactive dashboards for health metrics monitoring

C. Project Timeline

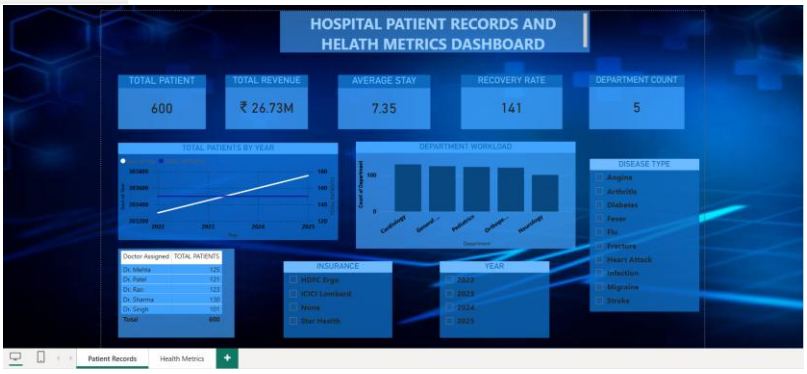
A simple table showing planned tasks and expected completion dates.

Phase	Task	Tentative Completion Date
1	Dataset collection, exploration, and understanding	*Week 1*
2	Data cleaning, preprocessing, and validation / Prompt design	*Week 2*
3	Analysis, AI tool integration (Copilot / Chatbot), and insight generation	*Week 2 – 3*
4	Dashboard / output build, testing, and review	*Week 3*
5	Report writing, GitHub upload, and video recording and submission	*Week 4*

7. Expected Outcomes / Conclusion

The Hospital Patient Records and Health Metrics Dashboard project demonstrates how healthcare data can be effectively organized, analyzed, and visualized to support better decision-making. By generating and preparing the dataset using **Microsoft Copilot** and developing interactive dashboards in **Microsoft Power BI**, the project successfully transforms raw patient records into meaningful insights.

The system enables real-time monitoring of key metrics such as admissions, discharges, bed occupancy, and treatment outcomes, reducing manual effort and improving reporting accuracy. Overall, the solution enhances operational efficiency, supports data-driven strategies, and contributes to improved patient care and hospital performance.



8. References

1. Healthcare Analytics Dashboard Tutorial – DataScientist.ca
   https://www.datascientist.ca/2025/11/10/health-care-analysis-power-bi-dashboard-step-by-step-tutorial-by-datascientist-ca/
2. Dashboard and analytics - https://youtu.be/TNtgJcIIg5Y
3. Microsoft Power BI Official Documentation
   https://learn.microsoft.com/power-bi/
   – Official documentation covering Power BI features such as data modeling, dashboards, and report creation.